Dirac Delta Function

The Dirac Delta Function

In many physical problems we want to model a source concentrated at a single point.

Example: a unit heat source at location ξ.

We represent this using the **Dirac delta**:

$$\delta(x - \xi)$$

which is zero everywhere except at $x = \xi$ and satisfies

$$\int_{-\infty}^{\infty} \delta(x - \xi)\, dx = 1.$$

It is best thought of not as an ordinary function but as an **idealized point mass**.

Sampling Property

The key property of the delta function is

$$\int_{-\infty}^{\infty} f(x)\,\delta(x-\xi)\,dx = f(\xi).$$

Thus the delta function "picks out" the value of f at ξ.

Heuristically we think of

$$\delta(x-\xi) = \begin{cases} 0 & x \neq \xi \\ \infty & x = \xi \end{cases}$$

in such a way that the total integral equals 1.

Delta as a Limit of Concentrated Functions

One way to understand the delta function is as a limit of ordinary functions whose mass concentrates at a point.

Example (Gaussian family):

$$\delta_\varepsilon(x) = \frac{1}{\sqrt{\pi}\varepsilon} e^{-x^2/\varepsilon^2}.$$

Then as $\varepsilon \to 0$

$$\int_{-\infty}^{\infty} \delta_\varepsilon(x)\, dx = 1 \text{ and } \delta_\varepsilon(x) \to \delta(x)$$

in the sense that

$$\int f(x)\delta_\varepsilon(x)\, dx \to f(0).$$

This viewpoint leads to the theory of **distributions**.

Test Functions

To make the delta function precise we introduce a class of "nice" functions called **test functions**.

Let

$$C_c^\infty(\mathbb{R})$$

denote the set of functions that

- are infinitely differentiable
- have **compact support**

Compact support means there exists an interval $[a, b]$ such that

$$\varphi(x) = 0 \quad \text{outside } [a, b].$$

Functions in $C_c^\infty(\mathbb{R})$ are called **test functions**.

Distributions and Integration by Parts

A **distribution** is a linear functional acting on test functions.

For example, the Dirac delta is defined by

$$\langle \delta, \varphi \rangle = \varphi(0), \qquad \varphi \in C_c^\infty(\mathbb{R}).$$

Derivatives of distributions are defined using integration by parts.

If T is a distribution, its derivative satisfies

$$\langle T', \varphi \rangle = -\langle T, \varphi' \rangle.$$

Example: Derivative of the Delta

Applying the definition to the Dirac delta:

$$\langle \delta', \varphi \rangle = -\langle \delta, \varphi' \rangle$$

gives

$$\langle \delta', \varphi \rangle = -\varphi'(0).$$

Thus δ' acts by differentiating the test function and evaluating at the origin.

This viewpoint explains many PDE identities, such as the derivative jump conditions that appear in Green's function constructions.